

Computer Networks Lab

Basic Router Access and CLI Configuration

NAME: Saqlain Mushtaq

ROLL NO: 2024-SE-35

SEMESTER: SE 4th

Lab: Lab 4

1. Access Privileged EXEC Mode:

Click on the router, go to the **CLI** tab, and press Enter to get started. Type the following command:

```
Router> enable
```

Explanation:

The enable command moves the router from User EXEC mode (where commands are highly limited) to Privileged EXEC mode. This higher-level mode is required before you can view system files or enter configuration modes.

2. Enter Global Configuration Mode:

```
Router# configure terminal
```

Explanation:

The configure terminal command shifts the router from Privileged EXEC mode into Global Configuration mode. This mode is necessary to make active changes that affect the overall operation of the router, such as setting passwords or configuring interfaces.

3. Configure Enable Password:

```
Router(config)# enable password cisco123
```

Explanation:

This command sets out a basic password required to enter Privileged EXEC mode, restricting unauthorized access. By default, this specific password is saved in plain text within the configuration file, making it vulnerable if the file is viewed.

4. Enable Password Encryption:

```
Router(config)# service password-encryption
```

Explanation:

This command applies a basic encryption algorithm (Type 7) to all current and future plain-text passwords stored on the router. It ensures that passwords, like the basic enable

password, are not easily readable to anyone looking over your shoulder at the configuration file.

5.Configure Enable Secret Password:

```
Router(config)# enable secret class456
```

Explanation:

This sets up a highly secure, heavily encrypted password for accessing Privileged EXEC mode. When both an enabled password and an enable secret are configured, the router will strictly demand the enable secret password, overriding the basic one.

6.Display Running and Startup Configurations:

First, exit back to Privileged EXEC mode, then view your configurations:

```
Router(config)# exit
Router# show running-config
Router# show startup-config
```

Explanation:

The show's running-config command displays the active, unsaved configuration currently running in the router's RAM. The show startup-config command displays the saved configuration stored in NVRAM, which dictates how the router will boot up.

7.Save the Configuration:

```
Router# copy running-config startup-config
```

Explanation:

This crucial command copies the active settings in RAM over to the non-volatile NVRAM. Saving the running-config to the startup-config ensures that all your password configurations will persist and load automatically even if the router loses power or reboots.